

ASIC Design Flow

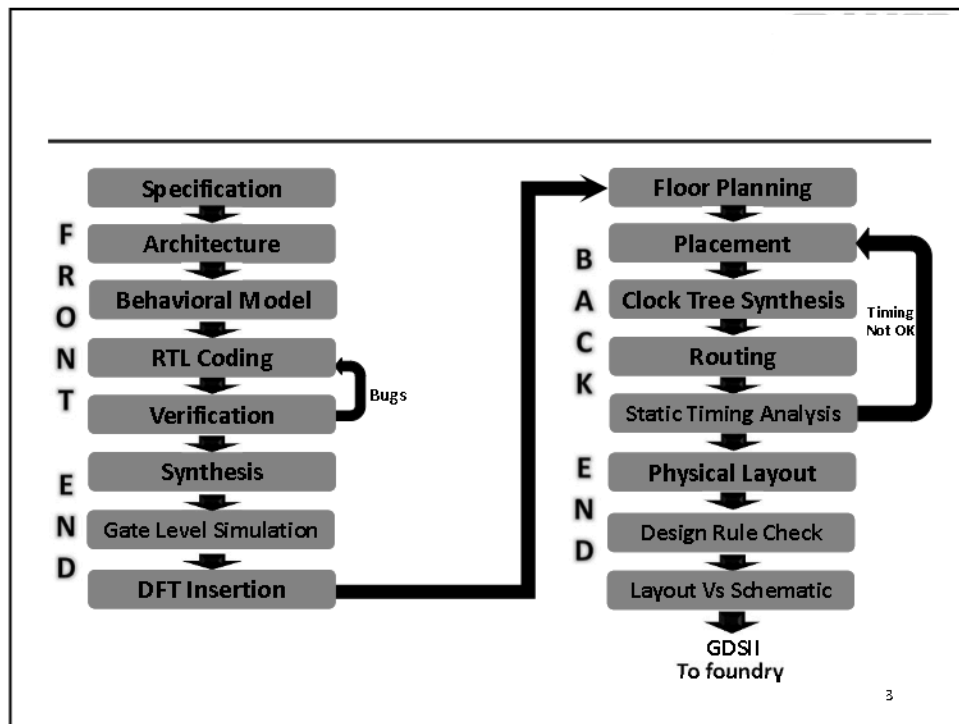
1

1

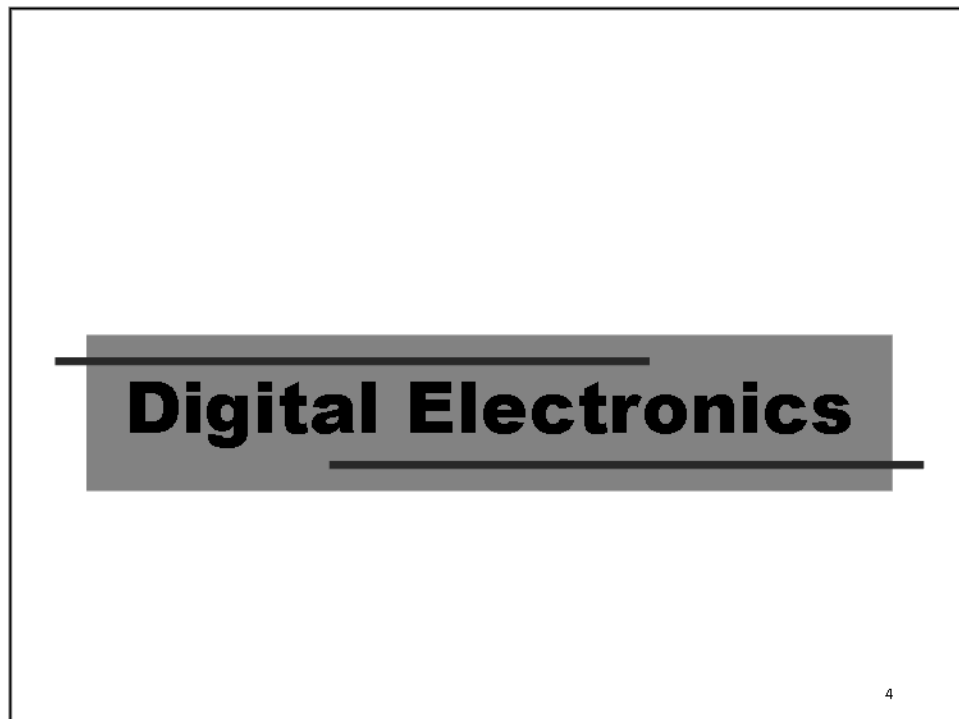
2

2

1



3



4

Course Agenda

- ♦ **Introduction**
- ♦ **Number Systems & Codes**
 - ♦ Conversion Techniques
 - ♦ Signed Numbers
 - ♦ Arithmetic Operations
 - ♦ Codes
- ♦ **Logic Circuits & Boolean Algebra**
 - ♦ Logic Gates
 - ♦ Boolean Algebra
 - ♦ NAND & NOR Circuits
 - ♦ Logic Minimization & K-Map
 - ♦ Don't care Conditions
- ♦ **Combinational Logic Circuits**
 - ♦ Design Procedure
 - ♦ Delays in combinational circuits
 - ♦ Adders & Subtractors
 - ♦ Encoders & Decoders
 - ♦ Multiplexer & Demultiplexer.
 - ♦ Implementation of Boolean functions using Multiplexers.
 - ♦ Tristate Buffers.

5

5

Course Agenda

- ♦ **Sequential Logic Circuits**
 - ♦ Latches
 - ♦ Flipflops
 - ♦ Delays in Sequential Circuits
 - ♦ Registers
 - ♦ *Buffer & Shift Registers*
 - ♦ Counters
 - ♦ *Classification*
 - ♦ *Modulus of a counter*
 - ♦ *Counter Design Process*
 - ♦ *Frequency division*
 - ♦ *Shift register Counters*
- ♦ **Finite State Machines**
 - ♦ Mealy Machine
 - ♦ Moore Machine
- ♦ **Glitches & Hazards**
 - ♦ Definition
 - ♦ Elimination of Hazards
- ♦ **Memories**
 - ♦ Classification of memories.
 - ♦ Basic Concepts of Memories
 - ♦ Composition of memories.

6

6

3

Number Systems

7

7

Agenda

- ♦ Binary, Decimal, Octal and Hexa Decimal Number
- ♦ Number system Conversions
- ♦ Signed Numbers
- ♦ Sign Magnitude Representation
- ♦ Complement Representation
- ♦ Arithmetic Operations using Complement number system.
- ♦ Codes

8

8

4

Number Systems

What is Number System?

- ♦ It can be defined as the Language of Numbers
- ♦ Each Language contains different set of symbols for representation of numbers.
- ♦ The most commonly used number systems are
 - ♦ Decimal Number System.
 - ♦ Binary Number System.
 - ♦ Octal Number System.
 - ♦ Hexa Decimal Number System.

Radix or Base

9

9

Number Systems

- ♦ Decimal Number System

{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }

Radix - 10

Example : 12495₍₁₀₎

- ♦ Binary Number System

{ 0, 1 }

Radix - 2

Example : 1111101₍₂₎

- ♦ Octal Number System

{ 0, 1, 2, 3, 4, 5, 6, 7 }

Radix - 8

Example : 71245₍₈₎

- ♦ Hexa Decimal Number System

{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F }

16

9ABD5₍₁₆₎

10

10

5

Number system conversions

Any radix to Decimal number system

No. of digits to the left of the radix point.

$$D = \sum_{i=-n}^{p-1} (d_i * r^i)$$

Value of the number

Radix of the number system

No. of digits to the right of the radix point.

11

11

Binary to Decimal Conversion

Example : $\overset{d_3 d_2}{1010}.\overset{d_1 d_0}{01}_{(2)}$

In this case,

$$D = \sum_{i=-n}^{p-1} (d_i * r^i) \Rightarrow D = \sum_{i=-2}^{4-1=3} (d_i * 2^i)$$

p = No. of digits to the left of the radix point = 4

n = No. of digits to the right of the radix point = 2

d = Value of the number

r = Radix of the number system = 2

$$\begin{aligned} D &= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) + (0 \times 2^{-1}) + (1 \times 2^{-2}) \\ &= 0.25 + 0 + 0 + 2 + 0 + 8 \\ &= 10.25_{(10)} \end{aligned}$$

12

12

Decimal to other number system

Integer part of Decimal number

Successive Division
by Required Radix number

Fractional part of Decimal number

Successive Multiplication
by Required Radix number

13

13

Decimal to other number system

Example : $125_{(10)} = ?_{(2)}$

Successive division

2	125	
2	62	- 1
2	31	- 0
2	15	- 1
2	7	- 1
2	3	- 1
	1	- 1

$$125_{(10)} = 1111101_{(2)}$$

14

14

Decimal to other number system

Example : $0.7_{(10)} = ?_{(2)}$

♦ Successive multiplication

$0.7 * 2 = 1.4$	$\Rightarrow 1$
$0.4 * 2 = 0.8$	$\Rightarrow 0$
$0.8 * 2 = 1.6$	$\Rightarrow 1$
$0.6 * 2 = 1.2$	$\Rightarrow 1$
$0.2 * 2 = 0.4$	$\Rightarrow 0$
$0.4 * 2 = 0.8$	$\Rightarrow 0$

$$0.7_{(10)} = 0.101100_{(2)}$$

$$125.7_{(10)} = 111101.101100_{(2)}$$

15

15

Binary to Octal

$(11101.011)_2 = (?)_8$

Step 1 : Create groups with 3 no. of binary digits from the fraction point.

11101.011

$\underbrace{011}_3$ $\underbrace{101}_5$ $\underbrace{011}_3$

Appending '0's' at MSB position of integer part, doesn't change the value of the number.

Step 2 : Represent each group with an equivalent octal number.

$$\Rightarrow (11101.011)_2 = (35.3)_8$$

16

16

Binary to Hexa

$$(11101.011)_2 = (?)_{16}$$

Step 1 : Create groups with 4 no. of binary digits from the fraction point.

$$\begin{array}{ccc} & & 11101.011 \\ \underbrace{0001} & \underbrace{1101} & \underbrace{0110} \\ 1 & D & 6 \end{array}$$

Appending '0's at MSB position of integer part & Appending '0's at LSB position in the fraction part of the number, doesn't change the value of the number.

Step 2 : Represent each group with an equivalent Hexa number.

$$\Rightarrow (11101.011)_2 = (1D.6)_{16}$$

17

17

Octal to Binary

$$(476.543)_8 = (?)_2$$

Step 1 : Represent each digit with a 3-bit binary equivalent

$$\begin{array}{ccccccc} \underbrace{4} & \underbrace{7} & \underbrace{6} & . & \underbrace{5} & \underbrace{4} & \underbrace{3} \\ 100 & 111 & 110 & & 101 & 100 & 011 \end{array}$$

$$\Rightarrow (476.543)_8 = (100\ 111\ 110.101\ 100\ 011)_2$$

18

18

Hexa to Binary

$$(ADE.54C)_{16} = (?)_2$$

Step 1 : Represent each digit with a 4-bit binary equivalent

$$\begin{array}{ccccccc} \underbrace{A} & \underbrace{D} & \underbrace{E} & . & \underbrace{5} & \underbrace{4} & \underbrace{C} \\ 1010 & 1101 & 1110 & & 0101 & 0100 & 1100 \end{array}$$

$$\Rightarrow (ADE.54C)_{16} = (1010\ 1101\ 1110.0101\ 0100\ 1100)_2$$

19

19

Signed Numbers

- ♦ Schemes to represent the signed numbers
 - Sign – Magnitude Representation
 - Complement Representation

20

20

10

Sign – Magnitude Representation

- ♦ MSB is used as sign bit & the other bits represents the magnitude.

0 – for Positive
1 – for Negative

- ♦ Example:

$1111_{(2)}$	=	15	⇒	Unsigned Number
$\underline{0}111_{(2)}$	=	+15	⇒	Signed <u>Positive</u> Number
$\underline{1}111_{(2)}$	=	-15	⇒	Signed <u>Negative</u> Number

- ♦ Range of the n bit signed magnitude integer :

$-(2^{(n-1)} - 1)$ to $+(2^{(n-1)} - 1)$

- ♦ Shortcomings:

- Complex Circuits
 - Two representations for 0
- 00000 +0
10000 -0

21

21

Complement Representation

- ♦ 1's complement - Complementing all bits

- ♦ Example :

$1111_{(2)}$ = +15₍₁₀₎
 $0000_{(2)}$ = -15₍₁₀₎

- ♦ 2's complement - Add 1 to 1's complement.

- ♦ Example :

$1111_{(2)}$ = +15₍₁₀₎
 $0001_{(2)}$ = -15₍₁₀₎

- ♦ Easy Method to get 2's Complement

$1011010_{(2)}$
 $0100101_{(2)}$ ⇒ 1's Complement
 $\quad \quad \quad +1$
 $0100110_{(2)}$ ⇒ 2's Complement

$1011010_{(2)}$
 $0100110_{(2)}$ ⇒ 2's Complement

Start from LSB and use the same digits until, you come across '1' and complement the remaining bits.

22

22

11

2's Complement for Binary Arithmetic

Addition

7	⇒	0 1 1 1
+4	⇒	+ 0 1 0 0
11	⇔	0 1 0 1 1

Subtraction

7	7	⇒	0 1 1 1
-4	+ (-4)	⇒	+ 1 1 0 0
3	⇔	1 0 0 1 1	X

Key Points

- ♦ For Subtraction, Take 2's Complement of the negative number and add it with the other number.
- ♦ Carry 1 indicates positive number
- ♦ Carry 0 indicates negative number.

4	4	⇒	0 1 0 0
-7	+ (-7)	⇒	+ 1 0 0 1
-3	⇔	1 1 0 1	

2's complement of 1101 is 0011

23

23

Complement Number system

System	Base	Possible complement systems
Binary	2	1's comp & 2's comp
Decimal	10	9's comp & 10's comp
Octal	8	7's comp & 8's comp
Hexa	16	15's comp & 16's comp
Unknown	n	(n-1)'s comp & n's comp

- ✓ n's complement of a number can be obtained by adding 1 to the (n-1)'s complement.
- ✓ (n-1)'s complement of a 'p' digit number can be obtained by subtracting the number from the maximum 'p' digit number.

24

24

12

Arithmetic using Complement System

Example :

$34_{(8)} - 27_{(8)}$
8's comp of $27_{(8)}$ is $(77 - 27) + 1 = 51$

34	$\Rightarrow$	34
-27	$\Rightarrow$	$+51$
$05_{(8)}$	$\Leftrightarrow$	105

'77' is the maximum 2 digit octal number

Additional Carry (if any) has to be discarded.

25

25

Arithmetic using Complement System

Example :

$2A_{(16)} - 4C_{(16)}$
16's comp of $4C_{(16)}$ is $(FF - 4C) + 1 = B4$

$2A$	$\Rightarrow$	$2A$
$-4C$	$\Rightarrow$	$+B4$
		$0DE$

'FF' is the maximum 2 digit hexa number

If the additional carry is not generated, It indicates that the result is a negative number.

16's comp of $DE_{(16)}$ is $(FF - DE) + 1 = 22$
So, $2A_{(16)} - 4C_{(16)} = -22_{(16)}$

To get the magnitude of result, 16's complement has to be calculated for the sum obtained.

26

26

13

Codes

- ♦ Symbolic representation
- ♦ Types of codes
 - ♦ Weighted binary codes.
 - ♦ Non-weighted codes.
 - ♦ Alphanumeric codes.
 - ♦ Error detecting codes.

27

27

Weighted Binary Codes

Decimal	8421	5421	2421
0	0000	0000	0000
1	0001	0001	0001
2	0010	0010	0010
3	0011	0011	0011
4	0100	0100	0100
5	0101	1000	1011
6	0110	1001	1100
7	0111	1010	1101
8	1000	1011	1110
9	1001	1100	1111

BCD Code

BCD code for $874_{(10)}$
 $874 = 1000\ 0111\ 0100_{(BCD)}$

28

28

14

Non – Weighted Codes

- ♦ Non –weighted codes are not positional weighted.
- ♦ Example :

Excess – 3 Code, Gray Code

Gray Code

- The no. of bit changes is only one between two consecutive numbers
- Used in low power applications

Decimal	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101

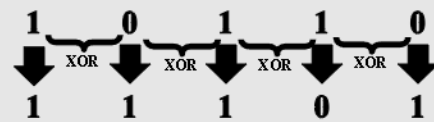
29

29

Binary and Gray code Conversion

- ♦ Binary to Gray code Conversion

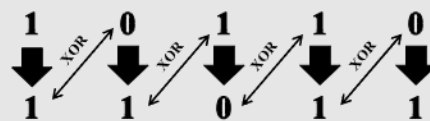
Convert $[10110]_2$ to gray code



$$10110_{(2)} = 11101_{(\text{gray})}$$

- ♦ Gray to Binary code Conversion

Convert $[10110]_{\text{gray}}$ to Binary code



$$10110_{(\text{gray})} = 11011_{(2)}$$

30

30

15

ASCII

- ♦ ASCII stands for American Standard Code for Information Interchange

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	EF	CR	SO	SI
1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
2		!	--	#	\$	%	&	.	{	}	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	:	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	/	]	^	_
6	.	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL

- ♦ The code uses 7 bits to encode 128 unique characters

31

31

Other codes

- ♦ Parity codes - Parity bits for detection of errors.
- ♦ Hamming code is for correcting the errors.
- ♦ Hollerith code.
- ♦ EBCDIC (Extended Binary Coded Decimal Interchange Code)
- ♦ ASCII Redefined with 8 bits to represent 256 different characters.

32

32

16